AARHUS UNIVERSITET

Prof. Akhil Arora

Natural Language Processing (NLP) – 520242U008
Midterm preparation – 2026
Suggested practice time: 42 minutes

Mock Exam 2026

Student ID : _____ Signature :

Do not turn the page before the start of the exam. This booklet is intended for double-sided printing and has 9 pages, the last ones possibly blank. Do not unstaple.

- This is a closed book exam. Non-programmable calculators are allowed. No other electronic devices of any kind are allowed.
- Place on your desk: your student ID, writing utensils, one double-sided A4 page cheat sheet if you have one; place all other personal items below your desk.
- This mock exam has **14 questions**. The suggested practice time is **42 minutes**.
- Each question has **6 options (A–F)** and is worth **one point**.
- Select **exactly one answer** per question. For statements labeled I, II, III, etc., select the option containing the **complete set of true statements**.
- An **answer sheet is included at the end of this booklet**. Record **all your answers on that sheet**. Only answers marked on the answer sheet will be graded. You can use the blank space in this booklet as scrap paper.
- Use a **black or dark blue ballpen** and clearly erase with **correction fluid** if necessary.
- If a question turns out to be wrong or ambiguous, we may decide to nullify it.
- **Do not tick** the boxes: **cross** them or **fill them in completely**. Do not write close to the boxes, and make sure an erased box is left **completely white**.

Observe these guidelines

Select an answer | Leave unselected | Correct an answer

What should **NOT** be done



Question 1

What is the primary purpose of backpropagation in neural networks?

A. To compute gradients of the loss function with respect to the input data.

B. To adjust the learning rate dynamically during training.

C. To directly update the weights while propagating error values backward through the network.

D. To compute gradients of the loss function with respect to the network's parameters.

E. To replace the forward pass when computing the network's predictions.

F. To automatically clear the optimizer's stored gradients after every parameter update.

Question 2

A model has perplexity 20 on dataset A and 40 on dataset B, using the same tokenizer and evaluation procedure. Perplexity is the exponential of the average negative log-likelihood per token, using natural logarithms.

I. The average negative log-likelihood per token is lower on A than on B.

II. Every observed token in A must receive exactly twice the probability of every observed token in B.

III. The average negative log-likelihood on B exceeds that on A by $\ln 2$ nats per token.

IV. The perplexity ratio alone does not establish a factor-of-two difference in downstream task performance.

Which option contains exactly the true statements?

A. I only.

B. I and II.

C. II and III.

D. I, II, and III.

E. III and IV.

F. I, III, and IV.

Question 3

Consider clipped PPO (Proximal Policy Optimization) in an RLHF pipeline.

I. Clipping discourages excessively large changes in action-probability ratios relative to the rollout policy.

II. Clipping guarantees that the KL divergence between the old and new policies never exceeds a fixed bound.

III. Using PPO makes the learned reward model exact and removes its estimation errors.

IV. An RLHF objective can use a separate KL penalty relative to a reference model in addition to PPO clipping.

Which option contains exactly the true statements?

A. I only.

B. I and IV.

C. II and III.

D. I, II, and IV.

E. I, III, and IV.

F. I, II, III, and IV.

Question 4

Consider standard Word2Vec without subword features.

I. It can produce a distinct learned embedding for a word outside its training vocabulary without any additional mechanism.

II. Standard CBOW and Skip-gram do not explicitly encode the relative positions of words within a fixed context window.

III. Arbitrarily shuffling each sentence leaves all local training co-occurrences, and therefore the learned representations, unchanged.

Which option contains exactly the true statements?

A. II only.

B. I and III.

C. I and II.

D. II and III.

E. I only.

F. I, II, and III.

Question 5

Which describes a key limitation of a standard RNN when its current hidden state summarizes a long sequence?

A. Its running time is always quadratic in sequence length, even with a fixed hidden dimension.

B. It must learn a different set of parameters for each timestep.

C. Its hidden-state updates must be linear, so it cannot learn nonlinear representations.

D. It cannot process sequences of different lengths with the same trained parameters.

E. It compresses the preceding context into a fixed-size hidden state, which can make retaining all relevant information difficult.

F. It computes every recurrent state independently, without access to an earlier hidden state.

Question 6

In a Transformer encoder-decoder model, what is the role of the causal mask in decoder self-attention?

A. It lets the decoder attend to all encoder outputs, aligning source and target sequences.

B. It randomly removes encoder outputs as a regularization mechanism.

C. It prevents a decoder position from attending to future target positions, preserving autoregressive prediction.

D. It removes all earlier target positions, allowing attention only to the current position.

E. It allows future target positions during training and blocks them only during inference.

F. It forces equal attention weights on every available target position.

Question 7

A training corpus contains the following six sentences. Split words by whitespace; do not add special tokens or form bigrams across sentence boundaries.

```
the cat sat on the mat
the dog chased after the cat
a cat and a dog ran
she sat on the sofa
they bought a new mat
the sofa was comfortable
```

There are 18 unique tokens and 32 tokens in total. Over this fixed vocabulary, use add-one Laplace smoothing:

$$P(A \mid B) = \frac{\text{count}(B, A) + 1}{\text{count}(B) + |V|}.$$

What are $P(\text{sat} \mid \text{on})$ and $P(\text{on} \mid \text{sat})$?

A. $P(\text{sat} \mid \text{on}) = 0$ and $P(\text{on} \mid \text{sat}) = 1$.
B. $P(\text{sat} \mid \text{on}) = \frac{1}{20}$ and $P(\text{on} \mid \text{sat}) = \frac{3}{20}$.
C. $P(\text{sat} \mid \text{on}) = \frac{3}{20}$ and $P(\text{on} \mid \text{sat}) = \frac{1}{20}$.
D. $P(\text{sat} \mid \text{on}) = \frac{1}{18}$ and $P(\text{on} \mid \text{sat}) = \frac{3}{18}$.
E. $P(\text{sat} \mid \text{on}) = \frac{2}{20}$ and $P(\text{on} \mid \text{sat}) = \frac{2}{20}$.
F. $P(\text{sat} \mid \text{on}) = \frac{1}{20}$ and $P(\text{on} \mid \text{sat}) = \frac{2}{20}$.

Question 8

Consider scaled dot-product self-attention, $\text{softmax}(QK^\top/\sqrt{d_k})V$, for a sequence of length n.

I. For each query, softmax normalizes the scores over key positions.
II. Without any mask, self-attention is automatically restricted to preceding positions.
III. Under the usual variance assumptions, division by $\sqrt{d_k}$ helps prevent excessively large scores and softmax saturation.
IV. The score matrix has n^2 entries, and its standard matrix multiplication costs $O(n^2 d_k)$ arithmetic operations.

Which option contains exactly the true statements?

A. I and III.
B. I, II, and III.
C. II and IV.
D. I, II, and IV.
E. I, II, III, and IV.
F. I, III, and IV.

Question 9

In a standard BART/T5 model, which representations supply the queries, keys, and values for a decoder block's encoder-decoder cross-attention? Learned projections are applied to these representations.

A. Queries: final encoder outputs; keys and values: decoder hidden states.

B. Queries, keys, and values: the same decoder hidden-state sequence.

C. Queries: decoder hidden states; keys and values: encoder input embeddings, bypassing the encoder layers.

D. Queries: decoder hidden states; keys and values: final encoder outputs.

E. Queries and keys: final encoder outputs; values: decoder hidden states.

F. Queries and keys: decoder hidden states; values: final encoder outputs.

Question 10

Which statement about these NLP models is correct?

A. T5 can perform generation tasks but cannot classify text by generating a class label.

B. Mean pooling of token embeddings without positional information can distinguish permutations of the same tokens.

C. A standard BERT encoder with its masked-language-model objective is not a left-to-right autoregressive language model.

D. Pretrained ELMo representations alone include a task-specific classification head and a complete sequence-generation system.

E. ELMo always supplies the same context-independent representation for a word, regardless of its surrounding sentence.

F. T5 generates each target token independently, without conditioning on earlier target tokens.

Question 11

This function should compute token-level perplexity from a non-empty list containing the probability assigned to each observed next token, with $0 < p_i \leq 1$. Use natural logarithms:

$$\mathrm{PPL} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\ln p_i\right).$$

```
import math

def perplexity(token_probabilities):
    n = len(token_probabilities)
    log_sum = sum(math.log(p) for p in token_probabilities)
    return math.exp(-log_sum)  # BUG: replace this line
```

Which replacement for the marked return line implements the formula for all valid inputs?

A. `return math.exp(-log_sum / n)`

B. `return math.exp(log_sum / n)`

C. `return math.exp(-log_sum) / n`

D. `return -log_sum / n`

E. `return math.exp(-log_sum * n)`

F. `return math.exp(-log_sum / (n + 1))`

Question 12

The classifier below should use the arithmetic mean of all token embeddings in each sequence. Each of the two sequences has five valid tokens, with no padding; the embedding dimension is eight. In `x`, the axes are batch, token position, and embedding coordinate, in that order.
`mean`, `sum`, and `max` reduce the specified dimension. For `max`, `.values` selects the maxima from its result.

```
import torch
from torch import nn

token_ids = torch.tensor([[1, 2, 3, 4, 5],
                          [5, 4, 3, 2, 1]])
embedding = nn.Embedding(10, 8)
classifier = nn.Linear(8, 3)

x = embedding(token_ids)
pooled = __I__
logits = classifier(pooled)
```

Which expression correctly fills `__I__` for any embedding weights?

A. `x.mean(dim=0)`

B. `x.mean(dim=2)`

C. `x.sum(dim=1)`

D. `x[:, 0, :]`

E. `x.mean(dim=1)`

F. `x.max(dim=1).values`

Question 13

The code below should implement causal self-attention. Row i represents a query position and column j a key position. A query may attend to positions $j \leq i$.
Broadcasting constructs a square Boolean mask. `masked_fill` replaces entries where the mask is `True`; softmax over the last dimension normalizes over keys.

```
import torch

length = 4
scores = torch.randn(length, length)
positions = torch.arange(length)

blocked = positions[None, :] < positions[:, None]  # BUG
masked_scores = scores.masked_fill(blocked, float("-inf"))
weights = torch.softmax(masked_scores, dim=-1)
```

I. The buggy condition masks keys at earlier positions, where $j < i$.
II. Replacing `<` with `>` in the marked line masks future keys and leaves the current position available.
III. Changing only the softmax dimension to 0 repairs the causal mask.
IV. Wherever future positions exist, the buggy mask leaves those future keys available to the query.

Which option contains exactly the true statements?

A. I only.
B. I and II.
C. I, II, and IV.
D. II, III, and IV.
E. I, III, and IV.
F. I, II, III, and IV.

Question 14

This PyTorch loop should make one ordinary optimizer update per minibatch, using only that minibatch's gradients. Gradient accumulation is not intended. The model, optimizer, scalar loss function, and at least two minibatches have already been defined.
`backward()` adds to stored gradients, `zero_grad()` clears them, and `step()` updates parameters without clearing their gradients.

```
model.train()
optimizer.zero_grad()

for inputs, targets in batches:
    logits = model(inputs)
    loss = criterion(logits, targets)
    loss.backward()
    optimizer.step()
```

Which single change corrects the unintended accumulation?

A. Move `optimizer.zero_grad()` inside the loop, immediately after `loss.backward()` and before `optimizer.step()`.

B. Swap `loss.backward()` and `optimizer.step()`.

C. Replace `loss.backward()` with `model.eval()`.

D. Move `optimizer.zero_grad()` to the start of each loop iteration, before the forward pass.

E. Add `model.train()` at the start of every iteration, leaving the existing gradient-reset call outside the loop.

F. Wrap `optimizer.step()` in `torch.no_grad()`, leaving the gradient-reset call outside the loop.



Mock Exam 2026

First name and last name:

.....

Student ID:

.....

Answer Sheet

Mark one answer per question.

Answers must be given exclusively on this sheet.
Answers given on the question sheets will be ignored.

Use a black or dark blue ballpen. Cross a box or fill it completely; do not tick. Leave any erased box completely white.

Question 1:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 2:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 3:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 4:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 5:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 6:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 7:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 8:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 9:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 10:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 11:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 12:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 13:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**
Question 14:	☐ **A**	☐ **B**	☐ **C**	☐ **D**	☐ **E**	☐ **F**

Total: 14 questions, 14 points.

Before finishing, check that you have transferred your answers to this sheet.